

$$1) a) \int_{-1}^0 \int_z^5 dx dy = \int_{-1}^0 x \Big|_z^5 dy = \int_{-1}^0 (5-z) dy = \int_{-1}^0 3 dy = 3y \Big|_{-1}^0 = 3(0 - (-1)) = 3 //$$

$$b) \int_z^4 \int_0^7 x^2 y dx dy = \int_z^4 y \frac{x^3}{3} \Big|_0^7 dy = \int_z^4 y \cdot \left(\frac{7}{3}\right) dy = \frac{1}{3} \frac{y^2}{2} \Big|_z^4 = \frac{1}{3} (8 - z) = 2 //$$

$$c) \iint_R y \ln x dA \quad \begin{cases} 2 \leq x \leq 3 \\ 1 \leq y \leq 2 \end{cases}$$

$$\int_1^2 \int_2^3 y \ln x dx dy = \int_1^2 y \cdot 3(\ln(3) - 1) - 2(\ln(2) - 1) dy = \frac{3}{2} (3 \ln(3) - 2 \ln(2) - 1) //$$

$$\textcircled{I} \int_2^3 \ln x dx = x(\ln x - 1) \Big|_2^3 \rightarrow 3(\ln(3) - 1) - 2(\ln(2) - 1)$$

$$\textcircled{II} \int_1^2 y dy = \frac{y^2}{2} \Big|_1^2 = \frac{2^2}{2} - \frac{1^2}{2} = 2 - \frac{1}{2} = \frac{4-1}{2} = \frac{3}{2}$$

$$u = \ln x \quad dv = dx$$

$$\frac{du}{dx} = \frac{1}{x}$$

$$dx = x du$$

$$v = x$$

* Integral por partes.

$$\int u dv = u \cdot v - \int v du$$

$$\ln x \cdot x - \int x \frac{dx}{x} = x \ln x - x = \boxed{x(\ln x - 1)}$$

$$\textcircled{2} a) \int_0^1 \int_0^y \sqrt{x+y} dx dy = \int_0^1 \frac{2}{3} y^{\frac{3}{2}} \cdot (2\sqrt{z^2-1}) dy = (2\sqrt{z^2-1}) \cdot \frac{2}{3} \int_0^1 y^{\frac{3}{2}} dy = \frac{2}{3} \cdot (2\sqrt{z^2-1}) \cdot \left(y^{\frac{5}{2}} \cdot \frac{2}{5} \Big|_0^1 \right) = \frac{2}{3} (2\sqrt{z^2-1}) \cdot \left(\frac{2}{5} \right) = \frac{4}{15} (2\sqrt{z^2-1}) //$$

$$\textcircled{I} \int_0^y \sqrt{x+y} dx = \int_0^y \sqrt{u} du = \int_0^y u^{\frac{1}{2}} du = \frac{2}{3} u^{\frac{3}{2}} \Big|_0^y = \frac{2}{3} (x+y)^{\frac{3}{2}} \Big|_0^y = \frac{2}{3} (2y)^{\frac{3}{2}} - \frac{2}{3} (y)^{\frac{3}{2}} = \frac{2}{3} 2\sqrt{z^2-1} (y)^{\frac{3}{2}} - \frac{2}{3} (y)^{\frac{3}{2}} = \frac{2}{3} y^{\frac{3}{2}} (2\sqrt{z^2-1}) //$$

$$u = x+y \quad \frac{du}{dx} = 1 \quad du = dx$$

$$b) \int_0^2 \int_{-y}^y (xy^2 + x) dx dy = \int_0^2 0 dy = 0 //$$

$$\textcircled{I} \int_{-y}^y (xy^2 + x) dx = \int_{-y}^y xy^2 dx + \int_{-y}^y x dx = y^2 \cdot \frac{x^2}{2} \Big|_{-y}^y + \frac{x^2}{2} \Big|_{-y}^y = \frac{x^2}{2} \Big|_{-y}^y (y^2 + 1) = \left(\frac{y^2}{2} - \frac{(-y)^2}{2} \right) (y^2 + 1) = 0 //$$

$$\textcircled{4} \iint_R 2x + 4y dA \quad \text{* Onde R é o região delimitada pelos eixos } x=1, y=x \text{ e } y=2x$$



$$\begin{cases} 0 \leq x \leq 1 \\ x \leq y \leq 2x \end{cases} //$$

$$\int_0^1 \int_x^{2x} 2x + 4y dy dx = \int_0^1 8x^2 dx = 8 \int_0^1 x^2 dx = 8 \cdot \frac{x^3}{3} \Big|_0^1 = \frac{8}{3}$$

$$\textcircled{I} \int_x^{2x} 2x + 4y dy = \int_x^{2x} 2x dy + \int_x^{2x} 4y dy = 2 \cdot x \cdot y \Big|_x^{2x} + 4 \cdot \frac{y^2}{2} \Big|_x^{2x} = 2x^2 + 6x^2 = 8x^2 + c //$$

③ $\iint_R 2xy \, dA$ *onde R delimitado pelas funções
 $y=0$, $y=x$, $x+y=2$



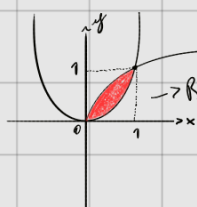
$$\begin{cases} 0 \leq y \leq 1 \\ y \leq x \leq 2-y \end{cases}$$

$$\iint_R 2xy \, dx \, dy = \int_0^1 \int_y^{2-y} 2xy \, dx \, dy = \int_0^1 -4y^2 + 4y \, dy = -4 \left(\frac{y^3}{3} \right) + 4 \left(\frac{y^2}{2} \right) \Big|_0^1 = -\frac{4}{3} + \frac{4}{2} = \frac{-8+12}{6} = \frac{4}{6} = \frac{2}{3} //$$

* Aqui temos que a região é delimitada por funções de y : $y=0$ (constante), $x=y$ e $x=2-y$

$$\textcircled{I} \int_y^{2-y} 2xy \, dx = 2y \left(\frac{x^2}{2} \Big|_y^{2-y} \right) = 2y \left(\frac{(2-y)^2}{2} - \frac{y^2}{2} \right) = 2y \left(\frac{4-4y+y^2}{2} - \frac{y^2}{2} \right) = y \cdot (4-4y) = -4y^2 + 4y$$

⑤ $\iint_R 2xy \, dA$ R é delimitado por $y=x^2$ e $y=\sqrt{x}$



$$\begin{cases} 0 \leq x \leq 1 \\ x^2 \leq y \leq \sqrt{x} \end{cases}$$

$$\iint_R 2xy \, dy \, dx = \int_0^1 \int_{x^2}^{\sqrt{x}} 2xy \, dy \, dx = \int_0^1 x^2 - x^5 = \left(\frac{x^3}{3} - \frac{x^6}{6} \right) \Big|_0^1 = \frac{1}{3} - \frac{1}{6} = \frac{2-1}{6} = \frac{1}{6}$$

$$\int_{x^2}^{\sqrt{x}} 2xy \, dy = 2x \cdot \left(\frac{y^2}{2} \Big|_{x^2}^{\sqrt{x}} \right) = 2x \cdot \left(\frac{x}{2} - \frac{x^5}{2} \right) = x^2 - x^5$$

* Temos a região entre funções de x : $y=x^2$ e $y=\sqrt{x}$

* Invertendo os eixos e as limitações

① $\begin{cases} 0 \leq y \leq 1 \\ y^2 \leq x \leq \sqrt{y} \end{cases}$ $\iint_R 2xy \, dx \, dy = \int_0^1 \int_{y^2}^{\sqrt{y}} 2xy \, dx \, dy = \int_0^1 y^2 - y^5 = \left(\frac{y^3}{3} - \frac{y^6}{6} \right) \Big|_0^1 = \frac{1}{3} - \frac{1}{6} = \frac{2-1}{6} = \frac{1}{6}$

$$\int_{y^2}^{\sqrt{y}} 2xy \, dx = 2y \cdot \left(\frac{x^2}{2} \Big|_{y^2}^{\sqrt{y}} \right) = 2y \cdot \left(\frac{y}{2} - \frac{y^5}{2} \right) = 2y \cdot \left(\frac{y - y^5}{2} \right) = y^2 - y^5$$

* Verificar com o professor qual está correto e qual não é o certo.

① Não aparece, mas está adaptado em relação ao problema proposto.

⑥ $\iint_R x^2 \, dA$ delimitado por $x=2$, $y=x+1$, $y=0$



$$\begin{cases} 0 \leq y \leq 3 \\ y-1 \leq x \leq 2 \end{cases} \iint_R x^2 \, dx \, dy = \int_0^3 \int_{y-1}^2 x^2 \, dx \, dy = \int_0^3 \left(\frac{x^3}{3} \Big|_{y-1}^2 \right) dy = \int_0^3 \left(\frac{8}{3} - \left(\frac{(y-1)^3}{3} \right) \right) dy = \frac{8}{3} y \Big|_0^3 - \frac{15}{12} = 8 - \frac{15}{12} = \frac{96-15}{12} = \frac{81}{12} = \frac{27}{4} //$$